Initialisation-dependent fixation in noisy weighted voter models

A resolution of Conjecture 2 of Ganesh, Hauert and Valla

Anonymous

Revised 27 October 2026

Abstract

Ganesh, Hauert and Valla (2025) studied a noisy weighted voter model for collective decisions in robot swarms. Their Conjecture 2 says that, on a complete graph, the probability that k agents holding the better option eventually win converges to the survival probability of a branching process. That process is built from the *equilibrium* distribution of the agents' signalling rates. The conjecture, however, specifies that the initial rates are *freshly sampled*. We show that this difference matters. When rates are bounded and every favourable rate lies strictly below every resident rate, we prove the population limit for arbitrary initial rate laws. It is governed by the transient mean resident rate, not by its equilibrium value. Freshly sampled residents update faster at first, because equilibrium over-represents slow residents (the inspection paradox). Whenever resident rates vary, this strictly raises the chance that the better option wins. The conjecture therefore fails as stated. Take favourable rates $1/2$ or 1 and resident rates 2 or 6, each with probability one half. The limit is then $0.7842\ldots$, not the conjectured $(3+\sqrt{37})/12 = 0.7569\ldots$. With a constant favourable rate 1 the limit is $0.70203\ldots$ instead of $2/3$. Under the equilibrium start that the source used in its simulations, the conjectured formula is exact for a constant favourable rate. For varied favourable rates the equilibrium-start limit is instead $1-(\mathbb{E}T_B/\mathbb{E}T_A)^k$, which again exceeds the conjectured value. Under the fresh start that the conjecture states and the equilibrium start that the source simulated, the stationary formula is conservative: it never overestimates the chance of reaching the better decision. (For other initial laws that over-represent slow residents it can overestimate; Section 4.2.) Its underestimate has a supremum of 27.34 percentage points, which no single model attains. If the fastest resident rate is at most three times the slowest, a proved ceiling is below 8.27 points (it tends to 8.27 as the slowest resident rate approaches the favourable rate, and is below 8 points when that ratio is at least 1.25). For uniformly spread rates the underestimate was below 3.5 points in every case we computed (slowest-resident-to-favourable ratio 1.25 or 2). We also give an explicit burn-in time after which the effect is negligible.

Keywords: weighted voter model; fixation probability; renewal process; inspection paradox; initialisation bias; branching process; swarm consensus.

1. Introduction

1.1. The question

In the weighted voter model, each agent advertises its preferred option for a random time and then copies the preference of a randomly chosen neighbour. Agents advertise better options for

longer, so better options are copied more often. Ganesh, Hauert and Valla (2025) analysed this rule as a decentralised algorithm for the best-of-n problem in swarm robotics (Valentini et al., 2017). In the *noisy* version, every agent that adopts an option measures its quality with error. The measured quality sets the mean of an exponential signalling period. Equivalently, the agent signals at a random *rate* R, and the mean signalling time is $T = 1/R$.

The source's Conjecture 2 concerns a large complete graph on which k agents initially prefer the better option, A, and the rest prefer B. It predicts the limiting probability that the population reaches consensus on A. The prediction comes from a branching process for the rare A agents. In that process, each A agent is copied at the rate λ at which the B population, *in equilibrium*, updates. The conjecture states that the initial measurements are taken freshly: the initial population starts "with initial values of T_A and T_B sampled as above" (Ganesh et al., 2025, p. 187).

1.2. Why the conjecture fails

The answer turns on a familiar fact about waiting times. Suppose half of the B agents sample a slow rate, 2, and half a fast rate, 6. Under fresh sampling, the mean B rate at time zero is 4. A slow agent, however, keeps its rate three times as long as a fast one. A snapshot of a population that has been running for a long time therefore over-represents slow agents: three quarters of the agents seen are slow. This is the inspection paradox, or length-biased sampling, of renewal theory. In equilibrium the mean B rate is only 3, which is the harmonic mean of the sampled rates. The fresh population relaxes from 4 to 3 as $3 + e^{-4t}$.

A rare A lineage is decided early. It either dies out within a few generations or grows large enough to be safe. During those first generations, freshly sampled B agents update, and hence copy A agents, faster than the equilibrium theory assumes. The early excess is never paid back. The rare favourable lineage therefore survives more often than the stationary branching process predicts. With a constant A rate 1 the corrected single-founder limit is $0.70203\ldots$ rather than $2/3$ (Section 4).

1.3. What the source itself simulated

The source authors were aware of a version of this issue. In their Section 4.7 they note that sampling from the steady state "is not the same as sampling from the site quality distributions" (p. 202–203). For their simulations they therefore drew the initial measurements from the size-biased equilibrium law, for agents of both opinions. Their reported agreement between simulation and Conjecture 2 is thus evidence about an equilibrium start, not about the fresh start that the conjecture states. Our results separate the two cases:

- With a constant favourable rate, the conjectured formula is *exactly right* under the equilibrium start the source simulated (equation (7)). It is *wrong* under the fresh start the conjecture states (Theorem 2).
- With varied favourable rates, the formula is wrong under both starts (Corollary 5 and Corollary 7). It is exact only for a hybrid start that neither the conjecture nor the simulations use: fresh favourable founders among equilibrium residents.

The source's Gamma-noise examples have overlapping supports of favourable and resident rates. They lie outside the class treated here (Section 5.5), so our results make no prediction about them.

We therefore describe our contribution as a resolution of Conjecture 2: we identify exactly when the stationary branching formula holds in the class we can treat, and what replaces it otherwise.

1.4. Main results

All results are for the complete graph, with exponential signalling periods, and with every favourable rate strictly below every resident rate (*separated supports*).

1. **Corrected limit (Theorems 1 and 4).** For arbitrary initial rate laws, the fixation probability converges to the survival probability of a *time-inhomogeneous* branching process. Its birth rate is the transient mean resident rate $b_\nu(t)$.
2. **Two counterexamples to Conjecture 2.** The first has varied favourable rates, so it does not rely on a degenerate quality law. With favourable rates 1/2 or 1 and resident rates 2 or 6, each with probability one half, the fresh-start limit is 0.7842.... The conjectured value is $(3+\sqrt{37})/12 = 0.7569...$, and the strict inequality is proved (Corollary 5). The second has a constant favourable rate 1 and the same residents. The limit is then 0.70203205392821... instead of 2/3 (Theorem 2 and equation (33)).
3. **Equilibrium start (Theorem 6, Corollary 7).** If all agents start from their equilibrium rate laws, as in the source's simulations, the limit is $1-(\mathbb{E}T_B/\mathbb{E}T_A)^k$. This is the formula of the noise-free model with qualities replaced by mean qualities. It exceeds the conjectured value whenever the favourable rates vary.
4. **How large the error can be (Theorem 3, Proposition 5.1, Table 3).** The single-founder underestimate is at most 27.3355... percentage points. This value is a supremum, approached only as the slowest resident rate approaches the favourable rate and the fastest resident rate grows without bound. No single model attains it. Capping the spread of resident rates reduces it (Table 3). If the fastest resident rate is at most three times the slowest, a proved ceiling is below 8.27 points; it tends to 8.27 as $b_0/a \to 1$ and is 7.77 at $b_0/a = 1.25$. Extreme two-point laws attain about 5 points (4.86 at $b_0/a = 1.25$, about 5.3 as $b_0/a \to 1$). Uniformly spread rates gave at most about 3.5 points in the cases computed ($b_0/a \in \{1.25, 2\}$). A ratio of 10 allows about 14 points for extreme two-point laws at $b_0/a = 1.25$, and more as $b_0/a \to 1$ (above 16 points attained; the ceiling tends to about 22.5).
5. **Burn-in (Section 6).** If the residents run alone for a time τ before the favourable agents appear, the error decays like $e^{-b_-\tau}$. We give an explicit sufficient burn-in time.

Table 1 collects the four possible starting conditions for the first counterexample.

Table 1: One model, four starts: favourable rates $1/2, 1$ and resident rates $2, 6$, each drawn with probability one half. "Equilibrium" means the length-biased law that a snapshot of a long-running population would show. Values with "..." after seven decimals come from a deterministic numerical integration that is not certified (Section 7.4); the closed forms are exact.

Favourable founders	Residents	Single-founder limit	Status
fresh	fresh (stated in Conjecture 2)	0.7842272...	Theorem 4; exceeds conjecture by Corollary 5

Favourable founders	Residents	Single-founder limit	Status
fresh	equilibrium	$(3+\sqrt{37})/12 = 0.7568969\ldots$	equals the conjectured formula (Lemma 7.3)
equilibrium	equilibrium (simulated in source §4.7)	$7/9 = 0.7777777\ldots$	Theorem 6 (exact)
equilibrium	fresh	$0.8030497\ldots$	Theorem 4

The practical message. Within the class treated here, the stationary branching formula is *conservative*. Under the fresh start that the conjecture states, and under the equilibrium start that the source simulated, the true probability of reaching the better decision is at least as large as the formula predicts. A swarm designer who used the formula was not led into overconfidence. A modeller who wants the formula to be *accurate* must, for a constant favourable rate, either start the simulation in equilibrium or let the residents burn in (Section 6). For varied favourable rates the equilibrium start is not enough (Corollary 7); the formula is exact only for equilibrium residents with fresh favourable founders.

1.5. The broader lesson

The mechanism is not special to swarms. A stationary approximation is often derived for a population in equilibrium and then applied from a fresh start. When individuals hold their state for random lengths of time, a snapshot of the equilibrium is length-biased, so slow individuals are over-represented (Pal et al., 2022, discuss the inspection paradox in a related setting). A fresh start is not length-biased. If the quantity of interest is decided early, as the fate of a rare lineage is, the transient matters at leading order. The same structure appears in four other literatures:

- *population genetics*, where the establishment probability of a new mutant depends on the environment during its first generations (Uecker and Hermisson, 2011). Simulations usually "burn in" a population before introducing a mutant. Section 6 quantifies such a burn-in.
- *epidemics on temporal networks*, where bursty, non-equilibrium activity changes early spreading (Karsai et al., 2011).
- *voter models with ageing or inertia*, in which the propensity to switch depends on the time since the last change (Stark et al., 2008; Fernández-Gracia et al., 2011). The reset-and-copy rule studied here is a renewal analogue.
- *simulation output analysis*, where initial-transient bias motivates warm-up deletion rules (White, 1997). Our bound (46) is a rare example of a provable warm-up length.

1.6. Relation to prior work

Several ingredients are classical:

- time-dependent linear birth–death processes (Kendall, 1948), which Uecker and Hermisson (2011) used to study fixation in changing environments;
- renewal monotonicity for decreasing-failure-rate laws (Brown, 1980);
- branching approximations for rare types and multitype branching processes (Athreya and Ney, 1972);
- law-of-large-numbers approximations for density-dependent Markov chains (Ethier and Kurtz, 1986);

- the effect of age structure and non-exponential generation times on invasion and fixation (Shpak, 2007; Hubbarde et al., 2007; Jafarpour et al., 2023). Alexandre et al. (2025) distinguish transient from asymptotic renewal quantities in a related population model.

We do not claim a new survival formula, a new renewal-monotonicity principle, or the first observation that initial age structure affects fixation. What is new is specific to the reset-and-copy model of Ganesh et al. (2025):

- the identification of the correct transient environment;
- a fixation bound that holds uniformly over all configurations and transfers rare-lineage survival to complete-graph fixation;
- the sharp error envelope for this model (Proposition 5.1 combines the classical renewal constant with a moment extremum; we have not compared it exhaustively with extremal renewal results);
- the equilibrium-start identity of Theorem 6, which is a short averaging calculation once the source's branching formula and the size-biased equilibrium law are written down. It is new for this model, but we do not claim it as a new branching-process theorem; related reproductive-value weightings appear in multitype branching theory.

The individual techniques (gambler's-ruin supermartingales, common-thinning couplings, finite-window branching approximations) are standard. Theorem 2 is essentially Brown's renewal monotonicity transferred to fixation through Proposition 3.6.

In the reset-and-copy model, an updating vertex copies an opinion and independently samples a new rate; it does not inherit the copied vertex's rate. The age-structured Moran model of Jafarpour et al. (2023) instead describes reproduction followed by random expulsion. The two models share branching methods but are different population processes.

1.7. Roadmap

- **Section 2** states the model, maps our notation to the source's, restates Conjecture 2, and gives the corrected limit for a constant favourable rate (Theorem 1).
- **Section 3** proves the population limit. It can be skipped on a first reading.
- **Section 4** proves that fresh starts strictly beat the stationary prediction (Theorem 2) and gives exact two-rate formulas and witnesses.
- **Section 5** asks how large the error can be: a sharp envelope (Theorem 3), the global supremum (Proposition 5.1), the error under rate caps (Table 3) and its detectability in simulation.
- **Section 6** gives the burn-in bound.
- **Section 7** treats random favourable rates: the general limit (Theorem 4), the strict fresh-start advantage (Corollary 5) and the equilibrium-start identity (Theorem 6 and Corollary 7).
- **Section 8** reports exact finite-population checks.
- **Section 9** discusses scope and open problems.

2. Model, notation and the corrected limit

2.1. The model

Let F_A and ν^A be probability laws on $[a_-, a_+]$, and let F and ν be probability laws on $[b_-, b_+]$, where

$$0 < a_- \le a_+ < b_- \le b_+ < \infty. \tag{1}$$

There are N vertices on the complete graph K_N, without self-loops. Vertex v holds an opinion $X_v \in \{A, B\}$ and a rate R_v. At rate R_v it updates: it chooses one of the other $N-1$ vertices uniformly at random and copies that vertex's opinion. It then draws a new rate, independently of everything else, from F_A if the new opinion is A and from F if it is B. The redraw happens even when the copied opinion equals the old one. This matches the source: an agent resamples its site "even if the opinion did not change" (Ganesh et al., 2025, p. 175). Initially k vertices hold A, where $k \ge 1$ is fixed as $N \to \infty$. Their rates are independent with law ν^A. The other $N-k$ vertices hold B with independent rates of law ν. Write $\alpha_k(N; \nu^A, \nu)$ for the probability of eventual consensus on A, called *fixation*.

The redraw-on-every-update rule is what drives the effect. If rates were redrawn only when the opinion changed, a resident copying another resident would keep its rate. While A is rare the resident rates would then stay essentially frozen at their initial values, so the initial mean rate would persist rather than relax. That variant is not the source's model, and we do not treat it.

When $F_A = \nu^A = \delta_a$ is a point mass we speak of a *constant favourable rate* and write $\alpha_k(N; \nu)$. A lower rate means longer signalling, so the favourable option A is the one whose rates lie below all resident rates.

2.2. Notation and the source's notation

The source writes measured qualities as signalling times T, with laws $\hat{F}_A, \hat{F}_B$, and rates $R = 1/T$, with laws F_A, F_B. We work with rates throughout. Table 2 gives the correspondence.

Table 2: Notation map.

This paper	Meaning	Ganesh et al. (2025)
F_A, F	redraw laws of favourable and resident rates	F_A, F_B (laws of $R_A = 1/T_A$, $R_B = 1/T_B$)
$\nu^A = F_A$, $\nu = F$	fresh start	"initial values … sampled as above" (Conjecture 2)
$\lambda = \left(\int r^{-1} F(dr)\right)^{-1} = 1/\mathbb{E}T_B$	harmonic-mean resident rate	λ, eq. (6)
$\mu(dr) = \lambda r^{-1} F(dr)$	equilibrium resident-rate law	G_B, eq. (5)
$m_A = \int x^{-1} F_A(dx) = \mathbb{E}T_A$	mean favourable signalling time	$\mathbb{E}[T_A]$
$G_A(dx) = m_A^{-1} x^{-1} F_A(dx)$	equilibrium favourable-rate law	$\hat{G}_A$ of Section 4.7, in rate form
$\nu = \mu$, $\nu^A = G_A$	equilibrium start	initialisation simulated in Section 4.7
c^*, $\pi = 1 - c^*$	stationary survival constant and extinction probability	c^*, π of Theorem 5, eq. (7)
$b_\nu(t)$	mean resident rate at time t from start ν	(not used)

2.3. Conjecture 2, restated

We restate the conjecture in our notation, clause by clause, with page references to the source.

- *Model (p. 186).* The noisy weighted voter model on K_N, as in Section 2.1. The phrase that an agent "signals the chosen opinion for the corresponding length of time" refers to the source's convention of p. 184: a measurement T starts an exponential signalling period with mean T.
- *Betterness (p. 186).* $\mathbb{E}[T_A] > \mathbb{E}[T_B]$, that is, $m_A > 1/\lambda$.
- *Initialisation (p. 187).* k agents hold A and $N-k$ hold B, "with initial values of T_A and T_B sampled as above". In our notation, $\nu^A = F_A$ and $\nu = F$.
- *Prediction (p. 187).* Let π be the extinction probability of a homogeneous multitype branching process. In it, each individual gives birth at rate λ, offspring types are drawn from F_A, and a type-x individual lives for an $\mathrm{Exp}(x)$ time. Theorem 5 of the source computes π for a founder whose type is drawn from F_A. The conjecture asserts $\lim_{N\to\infty} \alpha_k(N) = 1 - \pi^k$ for every fixed k.

Our class (1) satisfies every hypothesis: the graph is complete, and $m_A \geq 1/a_+ > 1/b_- \geq 1/\lambda$. A constant favourable rate is a point-mass quality law, which the conjecture does not exclude. Readers who regard that case as degenerate can rely instead on the two-point favourable law of Table 1.

Scope. All our theorems assume the complete graph, exponential signalling periods and separated supports (1). The source also analyses only K_N in this setting. General graphs, overlapping supports and non-exponential signalling are discussed in Section 9.

2.4. The resident process and the corrected limit

Define an autonomous *resident process* $R(t)$ on $[b_-, b_+]$ with generator

$$\mathcal{L}f(r) = r\left(\int f(s)F(ds) - f(r)\right), \qquad R(0) \sim \nu. \tag{2}$$

It redraws its rate from F at its current rate. Let

$$b_\nu(t) = \mathbb{E}_\nu R(t). \tag{3}$$

This is the mean update rate of a resident population that started from ν and has never met a favourable agent. For a constant favourable rate a, put

$$I_\nu(t) = \int_0^t [b_\nu(s) - a]\, ds, \qquad c_\nu = \left[1 + a\int_0^\infty e^{-I_\nu(t)}\, dt\right]^{-1}. \tag{4}$$

The integral converges because $b_\nu(t) \geq b_- > a$.

Theorem 1 (corrected fixation limit). Assume (1) with a constant favourable rate a. For every fixed $k \geq 1$,

$$\lim_{N\to\infty} \alpha_k(N;\nu) = 1 - (1 - c_\nu)^k. \tag{5}$$

The law μ of Table 2 is stationary for (2), so $b_\mu(t) = \lambda$ and $c_\mu = 1 - a/\lambda$. Theorem 1 then gives

$$\lim_{N\to\infty} \alpha_k(N;\mu) = 1 - (a/\lambda)^k. \tag{6}$$

For a constant favourable rate, the source's Theorem 5 gives $\pi = a/\lambda$, so (6) is exactly the conjectured value:

$$\text{under an equilibrium resident start, Conjecture 2's formula is correct.} \tag{7}$$

Under the fresh start $\nu = F$ it is not (Theorem 2).

2.5. The resident renewal equation

Let ν_t be the law of $R(t)$. Conditioning on the last redraw before t gives

$$\nu_t(dr) = e^{-rt}\nu(dr) + F(dr)\int_0^t e^{-r(t-s)}b_\nu(s)\,ds. \tag{8}$$

For a law η put $h_\eta(t) = \int re^{-rt}\eta(dr)$. Then

$$b_\nu(t) = h_\nu(t) + \int_0^t h_F(t-s)b_\nu(s)\,ds, \qquad \hat{b}_\nu(z) = \frac{\hat{h}_\nu(z)}{1-\hat{h}_F(z)}, \quad \hat{h}_\eta(z) = \int \frac{r}{z+r}\eta(dr). \tag{9}$$

Thus b_F is the ordinary renewal density of the mixture-of-exponentials signalling law. For a finite law $F = \sum_i f_i\delta_{r_i}$, the vector $x(t)$ of phase probabilities solves $x' = [fr^\top - \mathrm{diag}(r)]x$ with $x(0) = \nu$, and $b_\nu(t) = r^\top x(t)$.

3. Proof of the population limit

This section can be skipped on a first reading. We prove one result, Proposition 3.6, for a general favourable law F_A. Theorem 1 is its special case $F_A = \delta_a$, and Theorem 4 in Section 7 is the general case. The proof has two parts. The first is a fixation bound that holds from every configuration (Section 3.1). The second is a coupling showing that, over any fixed time window, the favourable agents behave exactly like a branching process (Sections 3.2–3.4). Section 3.5 joins the two.

Throughout, put

$$\rho = \frac{a_+}{b_-} < 1. \tag{10}$$

Let $M_N(t)$ be the number of favourable vertices, $\mathbf{M}_N(t)$ the multiset of their rates, and

$$S_N(t) = \sum_{v:\, X_v(t)=B} R_v(t)$$

the total resident rate. Conditional on the full state, M_N increases at rate β_N^+ and decreases at rate β_N^-, where

$$\beta_N^+ = \frac{M_N S_N}{N-1}, \qquad \beta_N^- = \frac{N-M_N}{N-1}\sum_{u:\, X_u=A} R_u. \tag{11}$$

Favourable vertices that copy a favourable vertex redraw their rate but leave M_N unchanged.

3.1. A fixation bound from every configuration

Lemma 3.1. Assume (1). From every state with $m \in \{1, \dots, N\}$ favourable vertices, whatever the rates,

$$\mathbb{P}(A \text{ fixes} \mid \text{current state}) \geq w_N(m) := \frac{1-\rho^m}{1-\rho^N} \geq 1-\rho^m. \tag{12}$$

Proof. For $0 < M_N < N$, we have $S_N \geq b_-(N-M_N)$ and every favourable rate is at most a_+. Hence

$$\beta_N^+ \geq b_- \frac{M_N(N-M_N)}{N-1}, \qquad \beta_N^- \leq a_+ \frac{M_N(N-M_N)}{N-1}, \tag{13}$$

so $\rho\beta_N^+ \geq \beta_N^-$. All jump rates are bounded by $2Nb_+$, so the process does not explode, and Dynkin's formula holds for bounded functions. The generator applied to $w_N(M_N)$ is

$$\beta_N^+[w_N(m+1)-w_N(m)] + \beta_N^-[w_N(m-1)-w_N(m)] = \frac{(1-\rho)\rho^{m-1}}{1-\rho^N}(\rho\beta_N^+ - \beta_N^-) \geq 0$$

at $m = M_N \in (0, N)$. It is 0 at the absorbing states. Hence $w_N(M_N(t))$ is a bounded submartingale.

Next we show that consensus occurs almost surely. While $0 < M_N < N$, the rate of changes of M_N is at least $\beta_N^+ \geq b_-$, because $M(N-M) \geq N-1$. It is also bounded above, so changes keep occurring. By (13), $\beta_N^+/(\beta_N^+ + \beta_N^-) \geq \vartheta := b_-/(b_- + a_+)$ at all times. Hence the next change is an increase with conditional probability at least ϑ. This follows from the formula $\mathbb{E}\int_0^\infty \beta^+(t)\exp(-\int_0^t(\beta^+ + \beta^-))dt$ for the probability that the first event of a two-type point process is of the first type. By the strong Markov property at successive change times, the next $N-1$ changes are all increases with probability at least ϑ^{N-1}, and that forces consensus on A. Repeating over disjoint blocks of $N-1$ changes, consensus fails to occur with probability at most $\lim_{\ell\to\infty}(1-\vartheta^{N-1})^\ell = 0$.

Finally, optional stopping for the bounded submartingale at the almost surely finite consensus time gives $\mathbb{P}(A \text{ fixes}) \geq w_N(m)$. □

3.2. A coupling lemma for jump processes

We compare two jump processes driven by the same Poisson random measures. The bound below is standard. We state it because the whole finite-time argument rests on it.

Lemma 3.2 (coupling by common thinning). Let $(\mathcal{G}_t)$ be a filtration and let $\Pi^1, \dots, \Pi^n$ be independent $(\mathcal{G}_t)$-Poisson random measures. Here Π^i lives on $[0,\infty)\times[0,\infty)\times\Theta_i$ with intensity $dt\,du\,\varrho_i(d\theta)$, and each ϱ_i is a probability law. Being $(\mathcal{G}_t)$-Poisson means that Π^i restricted to $[0,t]$ is $\mathcal{G}_t$-measurable and Π^i restricted to (t,∞) is independent of $\mathcal{G}_t$. Let X and Y be $(\mathcal{G}_t)$-adapted, right-continuous, piecewise-constant processes with values in a standard Borel space E, with finitely many jumps on bounded intervals and $X_0 = Y_0$. Suppose there are nonnegative $(\mathcal{G}_t)$-predictable processes $\ell_t^{X,i}, \ell_t^{Y,i}$ and maps $\Phi^{X,i}, \Phi^{Y,i} : \Omega\times[0,\infty)\times E\times\Theta_i \to E$ that are $\mathcal{P}\otimes\mathcal{B}(E)\otimes\mathcal{B}(\Theta_i)$-measurable, where $\mathcal{P}$ is the $(\mathcal{G}_t)$-predictable σ-field. We write $\Phi_t^{X,i}(x,\theta)$ for $\Phi^{X,i}(\omega,t,x,\theta)$. They have the following property. X changes only at atoms (t,u,θ) of some Π^i with $u \leq \ell_t^{X,i}$, and at *every* such atom (an accepted atom) X is assigned the image $X_t = \Phi_t^{X,i}(X_{t-},\theta)$, which

may equal X_{t-}. The same holds for Y with $\ell^{Y,i}$ and $\Phi^{Y,i}$. Let $\sigma = \inf\{t : X_t \neq Y_t\}$, and for $t \leq \sigma$ define the mismatch rate

$$m_t = \sum_{i=1}^{n} \int_{\Theta_i} \Big(\big|\ell_t^{X,i} - \ell_t^{Y,i}\big| + (\ell_t^{X,i} \wedge \ell_t^{Y,i})\, \mathbf{1}\{\Phi_t^{X,i}(X_{t-}, \theta) \neq \Phi_t^{Y,i}(X_{t-}, \theta)\} \Big) \varrho_i(d\theta).$$

(If $m_t = \infty$ the bound below is trivial.) Then for every stopping time τ,

$$\mathbb{P}(\sigma \leq \tau,\ \sigma < \infty) \leq \mathbb{E} \int_0^{\tau \wedge \sigma} m_t \, dt. \tag{14}$$

Proof. Let B_i be the set of (t, u, θ) with $t \leq \sigma$ such that either u lies strictly between $\ell_t^{X,i} \wedge \ell_t^{Y,i}$ and $\ell_t^{X,i} \vee \ell_t^{Y,i}$ (upper end included), or $u \leq \ell_t^{X,i} \wedge \ell_t^{Y,i}$ and the two maps disagree at (X_{t-}, θ). This set is $\mathcal{P} \otimes \mathcal{B}([0,\infty)) \otimes \mathcal{B}(\Theta_i)$-measurable: E is standard Borel, so its diagonal is measurable and σ is a stopping time; X_{t-} is predictable; and the maps Φ and thresholds ℓ are predictable by hypothesis. Both processes are piecewise constant, so the set $\{t : X_t \neq Y_t\}$ is a union of intervals closed on the left. Hence on $\{\sigma < \infty\}$ we have $X_{\sigma-} = Y_{\sigma-}$ and $X_\sigma \neq Y_\sigma$. Outside a null event on which two of the independent driving measures share an atom time, at time σ one of the processes changes, so there is a unique atom (σ, u, θ) of some Π^i. If that atom were outside B_i, then either it is accepted by neither process, so neither changes, or it is accepted by both and both are assigned the same image. Both contradict $X_\sigma \neq Y_\sigma$. So $\{\sigma \leq \tau, \sigma < \infty\}$ is contained in the event that $\sum_i \Pi^i(B_i \cap \{t \leq \tau\}) \geq 1$. Since B_i is predictable, the expectation of this count is $\mathbb{E} \int_0^{\tau\wedge\sigma} m_t \, dt$, and Markov's inequality gives (14). □

3.3. Construction of the population and its resident environment

We build the population from independent Poisson random measures, each with Lebesgue intensity in time and in the auxiliary coordinate u:

- for each vertex v, a *resident clock* Π_v on $[0,\infty) \times [0, b_+] \times [0,1]$;
- a *birth measure* Π^+ on $[0,\infty) \times [0,\infty) \times [0,1]^2$;
- a *favourable-update measure* Π^- on $[0,\infty) \times [0,\infty) \times [0,1]^3$;
- independent initial data.

Let F^{-1} and F_A^{-1} be quantile functions. Order favourable vertices by rate, breaking ties by label, so that a selection made with a uniform mark depends only on the multiset of favourable rates. Using left limits of all quantities, the dynamics are as follows.

- *(R)* At an atom (t, u, θ) of Π_v with v resident and $u \leq R_v(N - 1 - M_N)/(N-1)$, vertex v copies a resident. Its new rate is $F^{-1}(\theta)$.
- *(+)* At an atom $(t, u, \theta_1, \theta_2)$ of Π^+ with $u \leq \beta_N^+$, the mark θ_1 selects a resident v with probability R_v/S_N. That resident adopts A with rate $F_A^{-1}(\theta_2)$.
- *(−)* At an atom $(t, u, \theta_1, \theta_2, \theta_3)$ of Π^- with $u \leq \sum_{X_w = A} R_w$, the mark θ_1 selects a favourable vertex w with probability $R_w / \sum R$. If $\theta_2 \leq (N - M_N)/(N-1)$, vertex w copies a resident and becomes one, with rate $F^{-1}(\theta_3)$. Otherwise it copies a favourable vertex and stays favourable, with rate $F_A^{-1}(\theta_3)$.

Each transition of Section 2.1 then occurs at its correct rate. A resident v copies a resident at rate $R_v(N-1-M)/(N-1)$ and a favourable vertex at rate $\beta_N^+ R_v/S_N = R_v M/(N-1)$. A favourable vertex w copies a resident at rate $R_w(N-M)/(N-1)$ and a favourable vertex at rate $R_w(M-1)/(N-1)$. All rates are bounded, so this standard thinning construction defines the model (compare Ethier and Kurtz, 1986, Chapter 6). Let $(\mathcal{G}_t)$ be the filtration generated by the initial data and all the measures restricted to $[0,t]$.

Virtual residents. For each v, let $R_v^0(0) = R_v(0)$ if v starts as a resident; otherwise draw $R_v^0(0)$ from ν independently. At each atom (t,u,θ) of Π_v with $u \le R_v^0(t-)$, set $R_v^0(t) = F^{-1}(\theta)$. The processes R_v^0 are independent copies of the resident process (2).

Disturbed vertices. Call v *disturbed* by time t if, by then, either v has held A, or Π_v has had a *gap atom*. A gap atom is an atom (s,u,θ) with $R_v(s-)(N-1-M_N(s-))/(N-1) < u \le R_v(s-)$ while v was a resident. Let $H_N(t)$ be the number of disturbed vertices. An undisturbed vertex has been a resident throughout, and its rate equals its virtual rate. Below the lower threshold both rates jump to the same $F^{-1}(\theta)$; above $R_v = R_v^0$ neither jumps; and by definition there has been no atom in between. Every disturbed vertex has an actual rate and a virtual rate in $[0, b_+]$. Therefore

$$\left| S_N(t) - \sum_v R_v^0(t) \right| \le b_+ H_N(t). \tag{15}$$

Births occur at rate at most $b_+ M_N$, so $\mathbb{E}M_N(t) \le k e^{b_+ t}$ by Gronwall's inequality. Gap atoms occur at total rate at most $\sum_v R_v M_N/(N-1) \le 2b_+ M_N$. Hence

$$\mathbb{E}H_N(T) \le k + 3b_+ \int_0^T k e^{b_+ t} dt \le 3k e^{b_+ T}. \tag{16}$$

Lemma 3.3 (the resident environment). For every $T < \infty$,

$$\sup_{t \le T} \left| \frac{S_N(t)}{N-1} - b_\nu(t) \right| \longrightarrow 0 \quad \text{in probability and in } L^1. \tag{17}$$

Proof. Let $\overline{R}_N^0(t) = N^{-1} \sum_v R_v^0(t)$. Fix an integer L and grid points $t_j = j/L$.

1. At each of the finitely many grid points in $[0,T]$, $\overline{R}_N^0(t_j) \to b_\nu(t_j)$ in probability, by the law of large numbers for bounded independent variables.
2. Between grid points, each atom of some Π_v changes at most one virtual rate, by at most b_+. The number $C_{N,j}$ of such atoms in $(t_j, t_{j+1}]$ is Poisson with mean Nb_+/L. Hence $\sup_{u \in [t_j, t_{j+1}]} |\overline{R}_N^0(u) - \overline{R}_N^0(t_j)| \le b_+ C_{N,j}/N$, and $\max_j C_{N,j}/N \to b_+/L$ in probability.
3. By the forward equation, $b_\nu'(t) = \mathbb{E}\,\mathcal{L}\mathrm{id}(R(t))$, so $|b_\nu'| \le b_+(b_+ - b_-) \le b_+^2$.

Together, $\sup_{t \le T} |\overline{R}_N^0 - b_\nu| \le \max_j |\overline{R}_N^0(t_j) - b_\nu(t_j)| + b_+ \max_j C_{N,j}/N + b_+^2/L$. Its limit in probability is at most $2b_+^2/L$, and L is arbitrary. Finally, by (15),

$$\left| \frac{S_N}{N-1} - \overline{R}_N^0 \right| \le \frac{b_+ H_N(T) + b_+}{N-1}$$

for $t \le T$, and this tends to zero in L^1 by (16). The quantity in (17) is bounded by $3b_+$, so convergence in probability implies convergence in L^1. □

3.4. The finite-time branching approximation

Let $\mathcal{Z}$ be the following time-inhomogeneous branching process with types in $[a_-, a_+]$. At time 0 there are k individuals with independent types of law ν^A. A type-x individual dies at rate x. Every living individual gives birth at rate $b_\nu(t)$, and each child's type is drawn from F_A independently. Individuals never change type. Write $Z(t)$ for the population size and $\mathbf{Z}(t)$ for the multiset of types.

We build $\mathcal{Z}$ from the *same* measures $\Pi^\pm$ and the same founder types. At an atom of Π^+ with $u \le b_\nu(t)Z(t-)$, a child of type $F_A^{-1}(\theta_2)$ is born. At an atom of Π^- with $u \le \sum_{\mathbf{Z}} x$, the mark θ_1 selects an individual with probability proportional to its type, using the same ordering rule, and that individual dies. This gives $\mathcal{Z}$ its law: births occur at total rate $b_\nu Z$ with independent F_A types, and a type-x individual dies at rate x.

Proposition 3.4 (exact agreement on finite windows). For every $T < \infty$,

$$\mathbb{P}(\mathbf{M}_N(t) = \mathbf{Z}(t) \text{ for all } t \le T) \longrightarrow 1 \qquad (N \to \infty).$$

In particular, $M_N(T) \to Z(T)$ in total variation.

Proof. Apply Lemma 3.2 to $X = \mathbf{M}_N$ and $Y = \mathbf{Z}$, with values in finite multisets, driven by Π^+ and Π^-. The resident clocks do not change either multiset. Fix K, and let τ_K be the first time that M_N or Z exceeds K. While $\mathbf{M}_N(t-) = \mathbf{Z}(t-)$ and $t \le \tau_K$, the mismatch rates are as follows.

- Π^+. The thresholds are $M_N S_N/(N-1)$ and $M_N b_\nu(t)$. Both maps add a rate $F_A^{-1}(\theta_2)$. The mismatch rate is $M_N|S_N/(N-1) - b_\nu(t)| \le K|S_N/(N-1) - b_\nu(t)|$.
- Π^-. The thresholds coincide, since both equal the sum of the common multiset, and the selected rate coincides. The maps differ only when $\theta_2 > (N - M_N)/(N-1)$. Then the population keeps a favourable vertex, while $\mathcal{Z}$ loses an individual. The mismatch rate is at most $a_+ M_N(M_N - 1)/(N-1) \le a_+ K^2/(N-1)$.

By (14), with $\tau = T \wedge \tau_K$,

$$\mathbb{P}(\sigma \le T \wedge \tau_K) \le K\,\mathbb{E}\int_0^T \left|\frac{S_N(t)}{N-1} - b_\nu(t)\right| dt + \frac{a_+ K^2 T}{N-1}. \tag{18}$$

The right side tends to zero by Lemma 3.3.

Both M_N and Z have birth rate at most b_+ per individual. So $\sup_{t\le T} M_N(t)$ is at most k plus the number of births in $[0,T]$, whose mean is at most $k(e^{b_+ T} - 1)$, and similarly for Z. Markov's inequality gives

$$\mathbb{P}(\tau_K \le T) \le \frac{2ke^{b_+ T}}{K}. \tag{19}$$

Since $\{\sigma \le T\} \subseteq \{\tau_K \le T\} \cup \{\sigma \le T \wedge \tau_K\}$, the probability of any disagreement on $[0,T]$ has $\limsup_N$ at most $2ke^{b_+ T}/K$, for every K. □

The actual resident environment depends on the favourable counts. This causes no difficulty, because Lemma 3.2 needs only predictable intensities. Independence is used only for the virtual residents.

3.5. From survival to fixation

Lemma 3.5 (extinction or explosion). Assume $b_\nu \le b_+$ and $a_- > 0$. Almost surely, either $Z(n) = 0$ eventually or $Z(n) \to \infty$ as $n \to \infty$ through the integers.

Proof. Construct the process from independent exponential lifetime clocks and independent inhomogeneous Poisson birth streams, one per individual. Since the total birth rate is at most $b_+ Z(t)$, Z is dominated by a Yule process of rate b_+; hence it is non-explosive, $Z(n)$ exists for all n, and the (time, state) process is strong Markov. Fix $m_0 \ge 1$ and put $\varepsilon = [e^{-b_+}(1-e^{-a_-})]^{m_0} > 0$. Define integer stopping times by $\sigma_1 = \inf\{n \in \mathbb{N} : 1 \le Z(n) \le m_0\}$ and $\sigma_{\ell+1} = \inf\{n \ge \sigma_\ell + 1 : 1 \le Z(n) \le m_0\}$. Consecutive intervals $[\sigma_\ell, \sigma_\ell + 1]$ do not overlap.

Let $\ell \ge 1$. On $\{\sigma_\ell < \infty\}$, use the strong Markov property at σ_ℓ. With conditional probability at least ε, every individual alive at σ_ℓ dies within one time unit and none gives birth. Each individual's birth stream has rate at most b_+, and its death rate is at least a_-, independently across individuals. On that event the process is extinct at $\sigma_\ell + 1$ and $\sigma_{\ell+1} = \infty$. Hence $\mathbb{P}(\sigma_{\ell+1} < \infty) \le (1-\varepsilon)\mathbb{P}(\sigma_\ell < \infty)$, so $\mathbb{P}(\sigma_\ell < \infty \text{ for all } \ell) = 0$. Thus $Z(n) \in \{1, \dots, m_0\}$ for only finitely many n, almost surely. Taking the union over $m_0 \in \mathbb{N}$ proves the claim. □

Proposition 3.6 (fixation equals branching survival). Assume (1). Then

$$\lim_{N\to\infty} \alpha_k(N; \nu^A, \nu) = \mathbb{P}(Z(t) > 0 \text{ for all } t).$$

Proof. At any fixed time T, the favourable opinion cannot reappear once it has vanished. Lemma 3.1, applied conditionally on $\mathcal{G}_T$, bounds fixation from below. So

$$1 - \mathbb{E}\big[\rho^{M_N(T)}\big] \le \alpha_k(N; \nu^A, \nu) \le \mathbb{P}(M_N(T) > 0). \tag{20}$$

By Proposition 3.4 both bounds converge as $N \to \infty$, with $Z(T)$ in place of $M_N(T)$. Now let $T = n \to \infty$ through the integers. By Lemma 3.5, $\rho^{Z(n)} \to \mathbf{1}\{\text{extinction}\}$ almost surely, and $\mathbf{1}\{Z(n) > 0\} \to \mathbf{1}\{\text{survival}\}$. By dominated convergence, both bounds converge to $\mathbb{P}(\text{survival})$. The limits are taken in a fixed order: first $N \to \infty$ at fixed T, then $T \to \infty$. Lemma 3.1 is what makes this order legitimate. □

3.6. Proof of Theorem 1

With $F_A = \delta_a$, the process $\mathcal{Z}$ is a linear birth–death process with birth rate $b_\nu(t)$ and death rate a. Let $J(t) = a\int_0^t e^{-I_\nu(u)}du$. For one founder, its probability generating function is the classical formula of Kendall (1948):

$$G_t(w) = 1 - \frac{1-w}{[1+J(t)](1-w) + e^{-I_\nu(t)}w}, \qquad 0 \le w \le 1. \tag{21}$$

It can be checked directly in $\partial_t G_t(w) = (w-1)[b_\nu(t)w - a]\partial_w G_t(w)$ with $G_0(w) = w$. In particular $\mathbb{P}(Z(t) = 0) = G_t(0) = J(t)/(1+J(t))$. This increases to $1 - c_\nu$, because $J(\infty) < \infty$. Founders are independent, so Proposition 3.6 gives $\lim_N \alpha_k(N; \nu) = 1 - (1-c_\nu)^k$. □

4. Fresh starts and exact counterexamples

4.1. Fresh starts always help

Theorem 2 (strict fresh-start bias). Assume (1) with a constant favourable rate a. If F is not a point mass, then

$$b_F(t) > \lambda \quad \text{for every } t \geq 0, \qquad c_F > 1 - \frac{a}{\lambda}. \tag{22}$$

Consequently, for every fixed $k \geq 1$,

$$\lim_{N\to\infty} \alpha_k(N; F) > 1 - (a/\lambda)^k. \tag{23}$$

Proof. The equilibrium law μ is reversible for (2), because $\mu(dr)\, rF(ds) = \lambda F(dr)F(ds)$ is symmetric in (r, s). Hence $\mathcal{L}$ is a bounded self-adjoint operator on $L^2(\mu)$, with Dirichlet form

$$\langle f, -\mathcal{L}f\rangle_\mu = \frac{\lambda}{2} \iint [f(r) - f(s)]^2 F(dr)F(ds) = \lambda \operatorname{Var}_F(f) \geq 0. \tag{24}$$

Its kernel consists of the constants. Let $P_t = e^{t\mathcal{L}}$ and $\varphi(r) = r - \lambda$. Since $F(dr) = r\mu(dr)/\lambda$, $\int r\, \mu(dr) = \lambda$ and $\langle \varphi, 1\rangle_\mu = 0$,

$$b_F(t) = \frac{1}{\lambda}\langle \mathrm{id}, P_t\, \mathrm{id}\rangle_\mu = \lambda + \frac{1}{\lambda}\langle \varphi, P_t\varphi\rangle_\mu. \tag{25}$$

Let ϱ_φ be the spectral measure of $-\mathcal{L}$ associated with φ. By the spectral theorem,

$$\langle \varphi, P_t\varphi\rangle_\mu = \int_{[0,\infty)} e^{-\xi t}\, \varrho_\varphi(d\xi), \qquad \varrho_\varphi([0,\infty)) = \|\varphi\|_\mu^2. \tag{26}$$

If F is not a point mass, then $\varphi \neq 0$ in $L^2(\mu)$. The integrand is strictly positive, so (26) is strictly positive for every finite t. This proves $b_F(t) > \lambda$. Hence $I_F(t) > (\lambda - a)t$ for $t > 0$, and

$$a \int_0^\infty e^{-I_F(t)} dt < \frac{a}{\lambda - a},$$

which is $c_F > 1 - a/\lambda$. Theorem 1 gives (23). □

By (26), the excess $b_F(t) - \lambda$ is completely monotone in t. This agrees with classical decreasing-failure-rate renewal theory (Brown, 1980). The model-specific step is the conversion of that renewal fact into a strict fixation inequality through Theorem 1.

4.2. Two resident rates

Let $F = p\delta_{r_1} + (1-p)\delta_{r_2}$ with $a < r_1 < r_2$, and let $\nu = \theta\delta_{r_1} + (1-\theta)\delta_{r_2}$. Define

$$\kappa = (1-p)r_1 + pr_2, \qquad \lambda = \frac{r_1 r_2}{\kappa}, \qquad C_\theta = \theta r_1 + (1-\theta)r_2 - \lambda. \tag{27}$$

Solving the two-state resident chain gives

$$b_\nu(t) = \lambda + C_\theta e^{-\kappa t}, \qquad I_\nu(t) = (\lambda - a)t + \frac{C_\theta}{\kappa}(1 - e^{-\kappa t}). \tag{28}$$

Put $\omega = (\lambda - a)/\kappa$ and $\zeta = C_\theta/\kappa$. The substitution $y = e^{-\kappa t}$ gives the absolutely convergent series

$$c_\nu^{-1} = 1 + \frac{a}{\kappa} e^{-\zeta} \int_0^1 y^{\omega-1} e^{\zeta y} dy = 1 + \frac{a}{\kappa} e^{-\zeta} \sum_{j=0}^{\infty} \frac{\zeta^j}{j!(\omega + j)}. \tag{29}$$

The fresh start has $\theta = p$ and

$$C_p = \frac{p(1-p)(r_2 - r_1)^2}{\kappa} > 0. \tag{30}$$

The equilibrium start has $\theta = pr_2/\kappa$ and $C_\theta = 0$. Equations (28)–(29) also classify every other two-phase start: the sign of C_θ decides whether the limit lies above or below the equilibrium prediction. A start that over-represents slow residents, more than equilibrium does, makes the stationary formula an *overestimate*. The conservative direction is therefore a property of fresh and equilibrium starts, not of arbitrary ones.

4.3. Exact witnesses

Take $a = 1$, $r_1 = 2$, $r_2 = 6$ and $p = \theta = 1/2$. Then $\lambda = 3$, $\kappa = 4$, $C_p = 1$, and

$$b_F(t) = 3 + e^{-4t}. \tag{31}$$

The conjecture predicts the single-founder limit $2/3$. Theorem 1 instead gives

$$c_F = \left[1 + \frac{\sqrt{\pi}}{2} e^{-1/4} \operatorname{erfi}(1/2)\right]^{-1}, \qquad \operatorname{erfi}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{u^2} du. \tag{32}$$

Exact rational series bounds (Appendix A) certify

$$0.7020320539282127 < c_F < 0.7020320539282128. \tag{33}$$

The discrepancy is about 3.5365 percentage points. Its sign follows from Theorem 2 and does not depend on numerical evaluation.

A second witness, $a = 1$ and $F = \frac{1}{10}\delta_2 + \frac{9}{10}\delta_{100}$, has conjectured value $941/1000$. Its certified corrected limit lies between 0.9874094825209359 and 0.9874094825209361. A short argument also gives a fully rational proof of strict separation. On $[0, 1/10]$, $b_F(t) \geq b_F(1/10) > 38$, because $e^{-59/50} > 3/10$ by an alternating Taylor bound. So the limiting lineage dominates a homogeneous birth–death process with birth rate 38 and death rate 1 up to time $1/10$. After that, Lemma 3.1 applies with $\rho = 1/2$. Kendall's formula (21) then gives

$$\liminf_N \alpha_1(N; F) \geq \frac{37}{38 + 36e^{-37/10}} > \frac{370}{389} > \frac{941}{1000}, \tag{34}$$

using $e^{37/10} > 40$ from a positive Taylor sum. The verification script checks every rational inequality in this chain.

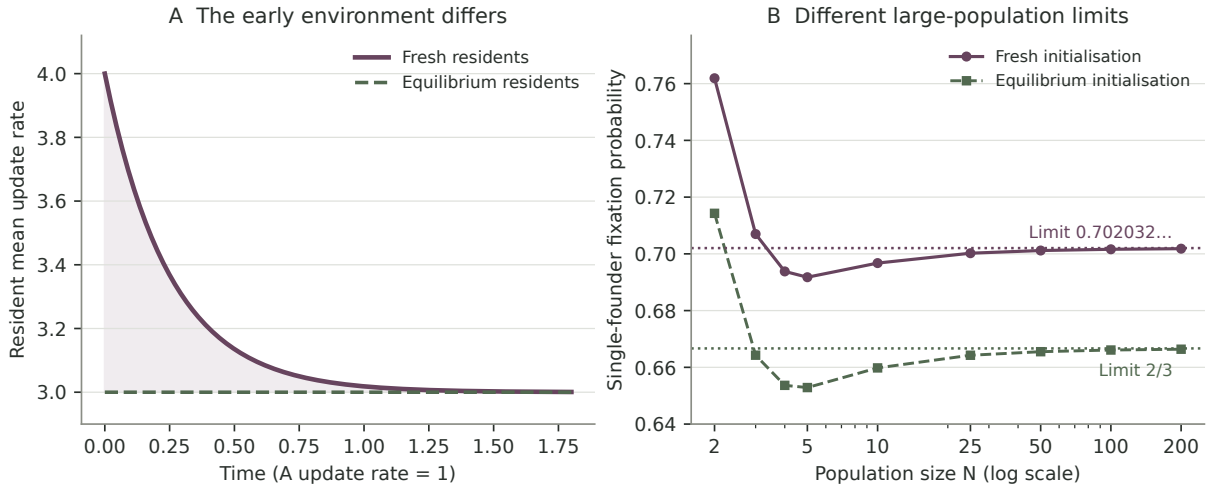


Figure 1: The constant-rate witness ($a = 1$, resident rates 2 and 6 with equal probabilities). Panel A: under a fresh start the mean resident rate starts at 4 and relaxes to its equilibrium value 3 as $3 + e^{-4t}$. Panel B: what to notice is that fresh-start fixation (upper, purple) approaches 0.70203, not the conjectured 2/3 (lower, green), and that the gap is already visible at $N \approx 10$. Points for $N \leq 5$ are exact rational solutions; larger N are floating sparse solves. Lines between points guide the eye; dotted lines mark the proved limits.

5. How large can the error be?

Throughout this section the favourable rate a is constant and the residents start fresh. Let b_0 be any number with $a < b_0 \leq b_-$, so that every resident rate is at least b_0. Put $\gamma = \lambda - a$.

5.1. Total excess intensity

Let $\chi(t) = b_F(t) - \lambda$ be the excess intensity, $D(t) = \int_0^t \chi$ its cumulative value, and $D = D(\infty)$. The spectral gap of $\mathcal{L}$ is at least b_0. To see this, note that $\mu \leq (\lambda/b_0)F$, so for every f, $\mathrm{Var}_\mu(f) \leq \mathbb{E}_\mu[(f - \mathbb{E}_F f)^2] \leq (\lambda/b_0)\,\mathrm{Var}_F(f)$. Together with (24) this gives $\langle f, -\mathcal{L}f\rangle_\mu \geq b_0\,\mathrm{Var}_\mu(f)$. Hence ϱ_φ is supported in $[b_0, \infty)$, and $\chi(t) \leq \chi(0)e^{-b_0 t}$. Writing $m_j = \mathbb{E}_F[R^{-j}]$ and expanding (9) at $z = 0$ gives $\widehat{h}_F(z) = 1 - m_1 z + m_2 z^2 + O(z^3)$ and $\widehat{b}_F(z) = 1/(m_1 z) + m_2/m_1^2 - 1 + O(z)$. Therefore

$$D = \frac{\mathbb{E}_F[R^{-2}]}{(\mathbb{E}_F[R^{-1}])^2} - 1, \tag{35}$$

which is the usual renewal-function constant. Since $R^{-2} \leq R^{-1}/b_0$,

$$0 \leq D \leq \frac{\lambda}{b_0} - 1. \tag{36}$$

Because $0 \leq D(t) \leq D$ and $I_F(t) = \gamma t + D(t)$, equation (4) yields

$$\frac{\gamma}{\lambda} \leq c_F \leq \frac{\gamma}{\gamma + ae^{-D}} \leq \frac{\gamma}{\gamma + ae^{-(\lambda/b_0 - 1)}}. \tag{37}$$

The first inequality is strict for non-degenerate F.

5.2. The sharp envelope

Theorem 3 (sharp envelope at fixed harmonic mean). Fix $0 < a < b_0 < \lambda$. Over all compactly supported resident-rate laws on $[b_0, \infty)$ with harmonic mean λ, the supremum of c_F is

$$\frac{\lambda - a}{\lambda - a + ae^{-(\lambda/b_0 - 1)}}. \tag{38}$$

The supremum is not attained.

Proof. The upper bound is (37). For $R > \lambda$ let

$$F_R = p_R \delta_{b_0} + (1 - p_R)\delta_R, \qquad p_R = \frac{b_0(R - \lambda)}{\lambda(R - b_0)}. \tag{39}$$

Each F_R has harmonic mean λ. In the notation of (27), $\kappa_R = p_R R + (1 - p_R)b_0 \to \infty$ and $C_{p_R}/\kappa_R \to \lambda/b_0 - 1$ as $R \to \infty$. Hence $I_R(t) \to \gamma t + \lambda/b_0 - 1$ for every $t > 0$. The integrands in (4) are bounded by $e^{-\gamma t}$, so dominated convergence gives the limit (38). For non-attainment, equality in $R^{-2} \le R^{-1}/b_0$ would force $R = b_0$ almost surely, contradicting $\lambda > b_0$. So $D < \lambda/b_0 - 1$ for every admissible law, and the last inequality in (37) is strict. □

The extremal family puts almost all of the excess intensity into a vanishingly short initial boundary layer. Very fast residents redraw almost at once, and a lineage that is still small at that moment profits from all of the excess. This, rather than the moment bound (36), is what makes the envelope sharp for fixation.

5.3. The global supremum

Time can be rescaled, so only λ/a matters. Put $\xi = \lambda/a - 1 > 0$ and

$$\Delta(\xi) = \frac{\xi}{\xi + e^{-\xi}} - \frac{\xi}{1 + \xi}. \tag{40}$$

Proposition 5.1 (global supremum). Over all constant favourable rates a and all resident laws whose support is a compact subset of the open interval (a, ∞) (so $\operatorname{supp} F \subset [b_0, \bar{b}]$ for some $a < b_0 \le \bar{b} < \infty$), the supremum of the single-founder underestimate $c_F - (1 - a/\lambda)$ equals $\Delta(\xi_*) = \max_{\xi > 0} \Delta(\xi)$. It is not attained.

Proof. Upper bound. Let $b_0 = \min \operatorname{supp} F > a$. If $\lambda = b_0$, then $F = \delta_{b_0}$ and the underestimate is 0. Otherwise, Theorem 3 gives strict inequality in (37). The right side of (37) increases with D, and $\lambda/b_0 - 1 < \lambda/a - 1 = \xi$, so $c_F < \gamma/(\gamma + ae^{-\xi})$. With $\gamma = a\xi$ this says $c_F - (1 - a/\lambda) < \Delta(\xi) \le \Delta(\xi_*)$.

Approach. Fix $\xi > 0$, and take $a = 1$ and $\lambda = 1 + \xi$. Choose $b_n \downarrow 1$ with $b_n < \lambda$. For each n, Theorem 3 provides a finite-rate law with harmonic mean λ, minimum rate b_n, and an underestimate within $1/n$ of $\gamma/(\gamma + e^{-(\lambda/b_n - 1)}) - \gamma/\lambda$. This quantity tends to $\Delta(\xi)$ as $n \to \infty$. Hence the supremum is at least $\Delta(\xi)$ for every ξ, and so at least $\Delta(\xi_*)$. □

The maximiser is unique. Let $\Omega(\xi) = (\xi e^{\xi/2} + e^{-\xi/2})/(1 + \xi)^{3/2}$. A direct computation shows that $\Delta'(\xi) > 0$ exactly when $\Omega(\xi) < 1$. The derivative of Ω has the sign of $\Phi(\xi) = (\xi^2 + 2)e^{\xi} - (\xi + 4)$. Since $\Phi(0) = -2$ and $\Phi'(\xi) = (\xi^2 + 2\xi + 2)e^{\xi} - 1 > 0$, the function Ω first decreases and then increases. As $\Omega(0) = 1$ and $\Omega(\xi) \to \infty$, it crosses 1 exactly once, at $\xi_* > 0$. Since $\Delta \to 0$ at both ends, ξ_* is the unique maximiser. Rational exponential bounds (Appendix A) certify

$$1.71891151345837 < \xi_* < 1.71891151345839,$$
$$0.2733554343215126 < \Delta(\xi_*) < 0.2733554343215181. \tag{41}$$

So the stationary approximation can underestimate single-founder fixation by amounts arbitrarily close to 27.3355 percentage points, but never by that much. The approach to the supremum is not uniform in the model. It needs $b_0 \downarrow a$, so the separation ratio $\rho = a/b_0$ tends to 1. It also needs the fastest resident rate to grow without bound. Both are physically extreme: real robots signal for a minimum duration, which caps the resident rate. Section 5.4 treats such caps. Finally, the result is about limits as $N \to \infty$; we do not claim a finite-N error bound.

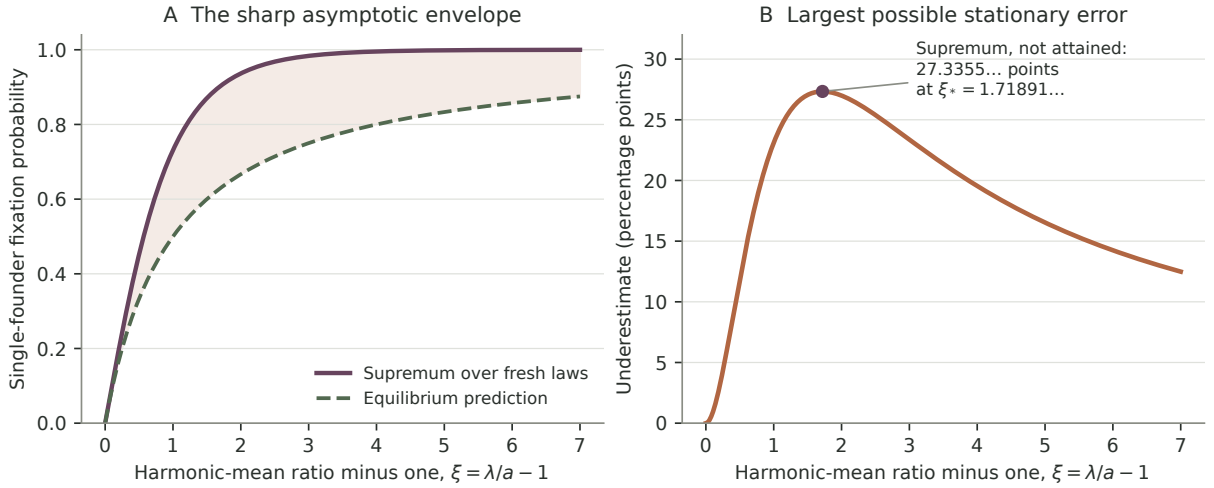


Figure 2: Panel A: the sharp envelope (38) with $b_0 \downarrow a$ (upper, purple), against the stationary prediction $\xi/(1+\xi)$ (lower, green), as functions of $\xi = \lambda/a - 1$. The upper curve is a supremum over models, not the fixation curve of any one model. Panel B: their difference, with its maximum of about 27.34 points at $\xi_* \approx 1.719$.

5.4. The error under a rate cap

Suppose that every resident rate also satisfies $R \le \bar{b} < \infty$. Then $X = 1/R$ lies in $[1/\bar{b}, 1/b_0]$. Taking expectations in $(X - 1/\bar{b})(1/b_0 - X) \ge 0$ gives the endpoint moment bound

$$D \le D(b_0, \bar{b}) := \frac{(\lambda - b_0)(\bar{b} - \lambda)}{b_0 \bar{b}}, \tag{42}$$

which is the Bhatia–Davis inequality in the variable X. Hence, by (37),

$$c_F - \frac{\gamma}{\lambda} \le \frac{\gamma}{\gamma + a e^{-D(b_0, \bar{b})}} - \frac{\gamma}{\lambda}. \tag{43}$$

Maximising the right side over $\lambda \in [b_0, \bar{b}]$ gives a *proved ceiling* for every law on $[b_0, \bar{b}]$. The bound (42) is attained by a law on the two endpoints. The ceiling (43) is not claimed to be sharp, because the timing of the excess intensity also matters. Table 3 compares the ceiling with values that are actually attained: the best law on the two endpoints, optimised over its weight, and the uniform law.

Table 3: Single-founder underestimate $c_F - (1 - a/\lambda)$, in percentage points, under a rate cap. The first two numeric columns are values attained by specific laws; the last is a proved upper bound over all laws on $[b_0, \bar{b}]$. Values are floating-point evaluations of the exact formulas (29), (4) and (43), computed by `stationary_identity_and_caps.py`. The global supremum 27.34 is approached only as $b_0/a \to 1$ and $\bar{b}/b_0 \to \infty$.

b_0/a	$\bar{b}/b_0$	Best two-point law (attained)	Uniform law on $[b_0, \bar{b}]$	Proved ceiling (43)
1.25	2	2.08	0.68	3.12
1.25	3	4.86	1.57	7.77
1.25	10	13.88	3.47	18.80
1.25	100	21.37	1.72	22.22
2	2	1.66	0.54	2.74
2	3	3.59	1.13	5.95
2	10	9.40	2.24	12.51
2	100	14.03	1.07	14.54

5.5. Could simulations have seen the effect?

The source estimates fixation probabilities from 1000 simulation runs per point (Ganesh et al., 2025, Section 4). Near $p = 0.7$, the standard error of such an estimate is $\sqrt{0.7 \times 0.3/1000} \approx 1.45$ percentage points, and a 95% interval has half-width about 2.8 points. The constant-rate witness has an asymptotic gap of 3.54 points, and at $N = 100$ the computed gap is 3.49 points (Table 5). This is about 2.4 standard errors. A dedicated two-sided test at the 5% level would detect it with probability only about two thirds. It would not be evident by eye on a plotted curve that also carries finite-N effects. There is a more basic reason why the source's simulations could not have detected it: they used the equilibrium start, for which (7) shows that the gap is zero when the favourable rate is constant. Their Gamma-noise examples also have overlapping supports, which lie outside our class.

6. Burn-in: pre-ageing the residents

Suppose that the residents evolve on their own for a time $\tau \geq 0$ before the k favourable agents arrive. Their initial law is then $\nu_\tau = FP_\tau$, so $b_{\nu_\tau}(t) = b_F(t + \tau)$. Let c_τ be the single-founder limit. Then $I_{\nu_\tau}(t) = \gamma t + \int_\tau^{\tau+t} \chi$. By (26), χ is strictly decreasing when F is non-degenerate, since $\chi'(t) = -\lambda^{-1} \int \xi e^{-\xi t} \varrho_\varphi(d\xi) < 0$. So $\int_\tau^{\tau+t} \chi$ strictly decreases in τ for each $t > 0$, and by (4) c_τ strictly decreases in τ. Its limit is the stationary value γ/λ.

The remaining excess is $D_\tau = \int_\tau^\infty \chi \leq De^{-b_0\tau}$, because ϱ_φ is supported in $[b_0, \infty)$. The argument of (37) gives

$$\frac{\gamma}{\lambda} \leq c_\tau \leq \frac{\gamma}{\gamma + a\exp(-D_\tau)}. \tag{44}$$

The function $y \mapsto \gamma/(\gamma + ae^{-y})$ has derivative $u/(1+u)^2$ with $u = (a/\gamma)e^{-y}$. That derivative is at most L, where $L = a\gamma/\lambda^2$ if $\gamma \geq a$ (then $u \leq 1$, where $u/(1+u)^2$ increases) and $L = 1/4$ otherwise. Hence

$$0 \le c_\tau - \frac{\gamma}{\lambda} \le LDe^{-b_0\tau}, \tag{45}$$

and for an absolute tolerance $\varepsilon > 0$ a sufficient burn-in time is

$$\tau \ge \frac{1}{b_0}\max\Big\{0,\ \log\frac{LD}{\varepsilon}\Big\}. \tag{46}$$

Bounds (45) and (46) concern the single-founder limit c_τ. For the k-founder fixation probability $p_k(c) = 1-(1-c)^k$, write $c_0 = \gamma/\lambda$, so $1-c_0 = a/\lambda$, and $\delta_\tau = c_\tau - c_0$. By the mean value theorem,

$$0 \le p_k(c_\tau) - p_k(c_0) = \Big(\frac{a}{\lambda}\Big)^k - \Big(\frac{a}{\lambda} - \delta_\tau\Big)^k \le k\Big(\frac{a}{\lambda}\Big)^{k-1} LDe^{-b_0\tau},$$

so a sufficient burn-in for tolerance ε in the k-founder probability is

$$\tau \ge \frac{1}{b_0}\max\Big\{0,\ \log\frac{k(a/\lambda)^{k-1}LD}{\varepsilon}\Big\}.$$

This reduces to (46) when $k = 1$.

If $D = 0$, the resident rate is deterministic and no burn-in is needed. For the witness of Section 4.3, $D = 1/4$, $b_0 = 2$ and $L = 2/9$. A tolerance of $\varepsilon = 10^{-3}$, that is 0.1 percentage point, in the single-founder probability then needs only $\tau \ge \frac{1}{2}\log(55.6) \approx 2.01$ mean favourable signalling times.

This bound has two uses. An experimenter who wants simulations to test the stationary theory should burn in the residents for this long. A swarm designer does not need to: in this model, a fresh start only helps the better option.

7. Random favourable rates

We now let the favourable rate law F_A on $[a_-, a_+]$ be arbitrary, with any initial favourable law ν^A. Section 7.1 identifies the limit. Section 7.2 treats the equilibrium start that the source simulated. Section 7.3 shows that fresh residents strictly help. Section 7.4 reports numerical values.

7.1. The limit

Let $\mathcal{Z}$ be the branching process of Section 3.4. For $s \le T$ and $x \in [a_-, a_+]$, let $e_s^{(T)}(x)$ be the probability that the lineage of a type-x individual alive at time s is extinct by time T. Let $\psi^{(T)}(t) = \int e_t^{(T)}(y)F_A(dy)$ be the same probability for a newborn at time t with a fresh type. More generally, for $z \in [0,1]$ let $Q_s^{(T,z)}(x) = \mathbb{E}_{x,s}[z^{Z(T)}]$ and $\psi_z^{(T)}(t) = \int Q_t^{(T,z)}(y)F_A(dy)$, so that $e^{(T)} = Q^{(T,0)}$.

Lemma 7.1 (backward equations). Write $\mathcal{E}_z(s,v) = \exp(-\int_s^v b_\nu(t)[1-\psi_z^{(T)}(t)]dt)$. Then

$$Q_s^{(T,z)}(x) = ze^{-x(T-s)}\mathcal{E}_z(s,T) + \int_s^T xe^{-x(v-s)}\mathcal{E}_z(s,v)\,dv. \tag{47}$$

Consequently, $s \mapsto Q_s^{(T,z)}(x)$ solves

$$\frac{d}{ds}Q_s(x) = x[Q_s(x)-1] + b_\nu(s)Q_s(x)\Big[1-\int Q_s(y)F_A(dy)\Big], \qquad Q_T(x) = z. \tag{48}$$

Proof. Types never change, so the remaining lifetime of the individual is $\mathrm{Exp}(x)$. Given that lifetime, its children are born at the points of a Poisson process of intensity b_ν, with independent F_A types. By the branching property, a child born at time t contributes an independent factor with mean $\psi_z^{(T)}(t)$. The probability generating functional of a Poisson process gives the factor $\mathcal{E}_z$. If the individual is alive at T it also contributes a factor z. Summing over death before or after T gives (47). Differentiating (47) in s gives $\partial_s Q_s = -x + (x + b_\nu(s)[1-\psi_z^{(T)}(s)])Q_s$, which is (48). □

For a finite favourable law $F_A = \sum_i \varpi_i \delta_{a_i}$, equation (48) is the finite system of backward equations that one integrates in practice. The extinction probabilities $e_s(x) = \lim_{T\to\infty} e_s^{(T)}(x)$ exist by monotonicity in T. By monotone convergence in (47) with $z = 0$,

$$e_s(x) = \int_s^\infty x e^{-x(v-s)} \exp\Big(-\int_s^v b_\nu(t)[1-\psi(t)]dt\Big)dv, \qquad \psi(t) = \int e_t(y)F_A(dy). \tag{49}$$

Theorem 4 (random favourable rates). Assume (1). For every fixed $k \geq 1$,

$$\lim_{N\to\infty} \alpha_k(N; \nu^A, \nu) = 1 - \Big[\int e_0(x)\,\nu^A(dx)\Big]^k. \tag{50}$$

Proof. Proposition 3.6 identifies the limit as the survival probability of $\mathcal{Z}$. Founder lineages are independent, and each goes extinct with probability $\int e_0\, d\nu^A$. □

Lemma 7.2 (a uniform survival bound). Let $\rho = a_+/b_-$. From any configuration of m individuals at any time, $\mathcal{Z}$ goes extinct with probability at most ρ^m.

Proof. Let Σ be the sum of the types of the living individuals, so $\Sigma \leq a_+ Z$. The generator of $\mathcal{Z}$ applied to ρ^Z is

$$b_\nu Z(\rho^{Z+1} - \rho^Z) + \Sigma(\rho^{Z-1} - \rho^Z) = \rho^{Z-1}(1-\rho)(\Sigma - \rho b_\nu Z) \leq \rho^{Z-1}(1-\rho)Z(a_+ - \rho b_-) = 0.$$

There is no explosion, since births are dominated by a Yule process. Localising at the exit times from $\{Z \leq K\}$ and letting $K \to \infty$ with Fatou's lemma, $\rho^{Z(t)}$ is a bounded supermartingale, so $\mathbb{E}\rho^{Z(n)} \leq \rho^m$. By Lemma 3.5, $\rho^{Z(n)} \to \mathbf{1}\{\text{extinction}\}$ almost surely, and bounded convergence gives $\mathbb{P}(\text{extinction}) \leq \rho^m$. □

7.2. The equilibrium start

Lemma 7.3 (equilibrium residents). If $\nu = \mu$, then $e_s(x) = e(x)$ does not depend on s, and

$$e(x) = \frac{x}{x + \lambda c^*}, \tag{51}$$

where $c^* \in (0,1)$ is the unique positive root of $\int \lambda(x+\lambda c)^{-1} F_A(dx) = 1$. In particular, fresh favourable founders give the limit $1-(1-c^*)^k = 1-\pi^k$, which is the conjectured value.

Proof. With $b_\mu \equiv \lambda$ the process is time-homogeneous. So $e_s^{(T)}$ depends only on $T-s$, and its limit e_s does not depend on s. Then (49) with constant ψ gives $e(x) = x/(x+\lambda c)$ with $c = 1-\psi$.

Integrating against F_A gives $c = \int \lambda c/(x+\lambda c)\, F_A(dx)$. By Lemma 7.2, $c \geq 1-\rho > 0$, so after division by c we get $\int \lambda(x+\lambda c)^{-1} F_A(dx) = 1$. The left side $g(c)$ is continuous and strictly decreasing in c, with $g(0+) = \lambda \int x^{-1} F_A(dx) = \lambda m_A > 1$ and $g(1) < 1$; so a root exists in $(0,1)$ and it is unique. Hence $c = c^*$, and (51) agrees with the source's Theorem 5 in our class. □

Theorem 6 (equilibrium start: the mean-quality formula). Assume (1), and start both opinions from equilibrium: $\nu = \mu$ and $\nu^A = G_A$, as in the source's Section 4.7 simulations. Then for every fixed $k \geq 1$,

$$\lim_{N\to\infty} \alpha_k(N; G_A, \mu) = 1 - \left(\frac{\mathbb{E}\, T_B}{\mathbb{E}\, T_A}\right)^k = 1 - \left(\frac{1}{\lambda m_A}\right)^k. \tag{52}$$

Proof. By Theorem 4 and Lemma 7.3,

$$\int e\, dG_A = \frac{1}{m_A} \int \frac{1}{x} \cdot \frac{x}{x+\lambda c^*} F_A(dx) = \frac{1}{\lambda m_A} \int \frac{\lambda}{x+\lambda c^*} F_A(dx) = \frac{1}{\lambda m_A},$$

where the last step is the defining equation of c^*. □

Theorem 6 says that, from equilibrium, the noisy model has the same large-population fixation probability as the noise-free model in which each option's quality equals its mean quality. In the noise-free model, single-founder fixation tends to $1 - q_B/q_A$, where q_A and q_B are the two qualities. The identity reflects reproductive value. A type-x individual has λ/x offspring on average, so the equilibrium weighting by $1/x$ is the reproductive-value weighting, and it makes the type-averaged extinction probability collapse.

Corollary 7 (Conjecture 2 under the simulated start). Under the hypotheses of Theorem 6, $1/(\lambda m_A) \leq \pi$, with equality if and only if F_A is a point mass. Hence, whenever the favourable rates vary, the equilibrium-start limit strictly exceeds the conjectured $1 - \pi^k$.

Proof. We have $\pi = \int e\, dF_A$ and $1/(\lambda m_A) = \int e\, dG_A = \mathbb{E}[e(X)X^{-1}]/\mathbb{E}[X^{-1}]$ with $X \sim F_A$. The function e is strictly increasing and $x \mapsto x^{-1}$ is strictly decreasing. Let X' be an independent copy of X. Then $\mathrm{Cov}(e(X), X^{-1}) = \frac{1}{2}\mathbb{E}[(e(X) - e(X'))(X^{-1} - X'^{-1})]$. This is negative unless $X = X'$ almost surely, that is, unless F_A is a point mass. □

For the example of Table 1, $\lambda = 3$ and $m_A = 3/2$, so the equilibrium-start limit is exactly $1 - 1/4.5 = 7/9$. The conjectured value is $(3+\sqrt{37})/12$. The stationary-identity script certifies both values in exact rational arithmetic. It also certifies the identity (52), and the strict inequality of Corollary 7, for three further favourable and resident laws.

7.3. Fresh residents strictly help

Corollary 5 (strict fresh-resident advantage). Assume (1) and that F is not a point mass. For every common initial favourable law ν^A and every $k \geq 1$, fresh residents ($\nu = F$) give a strictly larger limit than equilibrium residents ($\nu = \mu$). With $\nu^A = F_A$ this means that the fresh-start limit strictly exceeds the conjectured $1 - \pi^k$.

Proof. Let $\chi(t) = b_F(t) - \lambda > 0$, by (26). Build both branching processes on one space. Every individual carries independent ingredients:

- an $\mathrm{Exp}(x)$ lifetime;
- a *baseline* Poisson birth stream of rate λ;

- an *extra* Poisson birth stream of intensity $\chi(t)$ in absolute time;
- independent F_A types for the children of each stream.

In $\mathcal{Z}^F$, an individual's children are the points of both streams during its life. In $\mathcal{Z}^\mu$, they are the baseline points only, and only individuals descended from the founders through baseline births belong to $\mathcal{Z}^\mu$. Each process has the correct law: birth rates $\lambda + \chi = b_F$ and $\lambda = b_\mu$ respectively, with independent F_A types. The genealogy of $\mathcal{Z}^\mu$ is contained in that of $\mathcal{Z}^F$, so survival of $\mathcal{Z}^\mu$ implies survival of $\mathcal{Z}^F$.

For strictness, fix a founder of type x and some $T > 0$. Consider the event that the founder's lifetime lies in $(T, 2T)$, its baseline stream has no point in $(0, 2T)$, its extra stream has a point in $(0, T)$, and the subtree of the first extra child survives. On this event the founder has no children in $\mathcal{Z}^\mu$, whereas its lineage survives in $\mathcal{Z}^F$. The extra child's subtree has birth rate $b_F \geq b_-$ and death rates at most a_+. By Lemma 7.2 it survives with probability at least $1 - \rho$, independently of the founder's own ingredients given the birth time. Write e_0^F and e_0^μ for the extinction probabilities (49) under the two resident starts. Hence

$$e_0^\mu(x) - e_0^F(x) \geq (e^{-xT} - e^{-2xT}) e^{-2\lambda T} \Big(1 - e^{-\int_0^T \chi}\Big)(1 - \rho) > 0.$$

The restriction applies only to the founder's own streams; all descendants' streams are unrestricted. Averaging over ν^A preserves strictness, and (50) proves the claim for every k. □

7.4. Numerical values and a computable bracket

For finite favourable laws, (48) can be integrated numerically. Lemma 7.2, conditioned at time T, and the fact that ultimate survival requires survival at T give the bracket

$$1 - \Big[\int Q_0^{(T,\rho)}\, d\nu^A\Big]^k \leq \lim_{N\to\infty} \alpha_k \leq 1 - \Big[\int Q_0^{(T,0)}\, d\nu^A\Big]^k. \tag{53}$$

By Lemma 3.5, both sides converge to the limit as integer $T \to \infty$. The inequalities (53) are proved. The *values* we report from them are numerical, not certified: we integrate (48) with an ordinary floating-point solver (SciPy's DOP853, cross-checked with Radau), which supplies no rigorous error bound. For the example of Table 1, with fresh residents and fresh favourable founders, the numerical endpoints of (53) are 0.7841882234 and 0.7842430477 at $T = 4$, and 0.7842272885 and 0.7842272947 at $T = 8$.

Exact finite-population calculations support Theorems 4 and 6. Table 4 reports the probability that a single favourable founder fixes in the count chain with two favourable and two resident rates. For $N \leq 5$ the values are exact rational solutions; for larger N they are floating sparse solves with recorded residuals below 10^{-9}.

Table 4: Single-founder fixation in the finite population for the example of Table 1, under the four starts. The last row gives the limits: exact closed forms where available, otherwise the numerical value of (53) at $T = 12$.

N	fresh / fresh	fresh founders, equilibrium residents	equilibrium / equilibrium	equilibrium founders, fresh residents
2	1108/1365	703/910	1081/1365	3392/4095
3	0.7804688061	0.7465156976	0.7673674466	0.7989468709
4	0.7766495256	0.7449666725	0.7660039698	0.7954392892
5	0.7774833222	0.7469228746	0.7679767124	0.7963395208
10	0.7813643381	0.7526212169	0.7736076748	0.8002278429
25	0.7832269088	0.7553742164	0.7762973435	0.8020667952
50	0.7837472904	0.7561624295	0.7770642809	0.8025784494
100	0.7839920281	0.7565360172	0.7774273437	0.8028187952
∞	0.7842272930	$(3+\sqrt{37})/12 =$ 0.7568968775	$7/9 =$ 0.7777777778	0.8030496711

8. Deterministic verification of the constant-rate witness

For two resident rates and a constant favourable rate, the finite population reduces exactly to states (m, j). Here m is the number of favourable vertices and j the number of slow residents among the other $n = N - m$. Write $r_1 < r_2$ for the slow and fast rates and p for the probability of drawing the slow rate. The state-changing rates are:

New state	Rate
$(m-1, j+1)$	$amnp/(N-1)$
$(m-1, j)$	$amn(1-p)/(N-1)$
$(m+1, j-1)$	$r_1 jm/(N-1)$
$(m+1, j)$	$r_2(n-j)m/(N-1)$
$(m, j-1)$	$r_1 j(n-1)(1-p)/(N-1)$
$(m, j+1)$	$r_2(n-j)(n-1)p/(N-1)$

The verification script solves the resulting harmonic equations exactly in rational arithmetic for $N = 2, 3, 4, 5$. Every equation has zero exact residual. A separate implementation enumerates labelled vertex configurations for $N = 2, 3, 4$, and its floating results agree with the exact values. Larger populations are solved by deterministic sparse linear algebra, with residuals recorded.

For $a = 1$ and $F = (\delta_2 + \delta_6)/2$, the exact fresh-start probabilities are 16/21, 2864/4051, 443968/639901 and 310846319872/449354284245 for $N = 2, 3, 4, 5$. Their initial decrease shows that convergence need not be monotone in N. Larger values, rounded to ten decimals, are:

Table 5: Constant-rate witness: finite-N fixation probabilities. Floating values are rounded to ten decimals; the eleventh and twelfth decimals can vary by one unit across NumPy/SciPy versions.

N	Fresh residents	Gap to limit	Equilibrium residents	Gap to 2/3
10	0.6967465143	5.29×10^{-3}	0.6597969966	6.87×10^{-3}
25	0.7002259952	1.81×10^{-3}	0.6642598910	2.41×10^{-3}
50	0.7011711989	8.61×10^{-4}	0.6655113963	1.16×10^{-3}
100	0.7016113989	4.21×10^{-4}	0.6661003029	5.66×10^{-4}
200	0.7018240813	2.08×10^{-4}	0.6663862149	2.80×10^{-4}
∞	0.7020320539		2/3	

Doubling N from 100 to 200 roughly halves both gaps, which is consistent with an $O(1/N)$ rate. We do not prove such a rate. The finite-N values support the construction, but they do not prove the limit theorem. The decimal intervals for the limits and for the global maximiser are separately certified by exact rational arithmetic (Appendix A).

A basic normalisation check is the deterministic resident law $F = \delta_b$. The fresh and equilibrium laws then coincide, $b_F(t) = b$, equation (4) gives $c = 1 - a/b$, and Theorem 1 reduces to the usual biased-voter fixation limit. Multiplying all rates by a positive constant only rescales time and leaves every fixation formula unchanged.

9. Discussion

What is resolved. Within the class (1) on complete graphs, Conjecture 2 holds exactly when residents start in equilibrium and favourable founders start fresh. That hybrid start is used by neither the conjecture nor the source's simulations. Under the stated fresh start the conjecture is false, whenever resident rates vary, by Corollary 5 and Theorem 2. Under the simulated equilibrium start it is true for a constant favourable rate, by (7). It is false for varied favourable rates, by Corollary 7, where the correct limit is the mean-quality formula of Theorem 6. Under both the fresh and the equilibrium start, the conjectured value is an underestimate. The source's Theorem 5, a calculation of extinction in its stated branching process, is untouched. What fails is the identification of fresh finite populations with that stationary process.

Fair reading of the source. One can argue that the conjecture was *meant* to describe the equilibrium regime and that "sampled as above" is a drafting slip. Section 4.7 of the source supports that reading. Even then our results matter in two ways. First, for varied favourable rates the conjectured formula is not the equilibrium answer either (Corollary 7). Second, the fresh start is the natural start for a swarm that begins a decision, and the paper quantifies how much it helps.

The general lesson. Equilibrium approximations are routinely applied to systems that start fresh. When individuals hold their states for random times, the equilibrium snapshot is length-biased. A quantity decided in the early transient, such as the survival of a rare lineage, then differs at leading order from its stationary prediction. Burn-in removes the difference, at an exponential rate set by the spectral gap (Section 6).

Open problems.

1. *Overlapping supports.* If some favourable rates exceed some resident rates, the uniform comparison of Lemma 3.1 fails, and we have no substitute bridge from early survival to fixation. This is the generic case for noisy measurements of nearby qualities, including the source's Gamma-noise examples. We expect the transient mechanism to persist there, but we have not proved it.
2. *Non-complete graphs.* On sparse, regular or geometric graphs, the early-transient mechanism should persist locally, but the limit formula will not carry over directly. The source also leaves general graphs open.
3. *A sharp envelope under a rate cap.* Table 3 brackets the worst case between attained values and the ceiling (43). The exact extremum over laws on $[b_0, \bar{b}]$ is an optimal-timing problem that we have not solved.
4. *A finite-N rate.* Tables 4 and 5 suggest an $O(1/N)$ approach to the limit. An explicit bound would make the corrections usable for swarms of tens to hundreds of agents.
5. *Non-exponential signalling.* For signalling laws with increasing failure rate, the fresh-start renewal intensity can lie below its equilibrium value at first. The direction of the bias could then reverse.

Declarations

Author. Anonymous.

Version and status. Version 1.0.0-candidate, an unrefereed research candidate. DOI: https://doi.org/10.5281/zenodo.23687032.

AI provenance. This manuscript was produced with AI systems based on large language models. They drafted the text and proofs, wrote the verification scripts, and carried out the literature and citation checks. The revision responds to five internal AI review reports and one review by a model from a different family. The proofs have not been formalised in a proof assistant or refereed by human experts. No human author takes responsibility for the mathematics; readers should treat the results as a research draft to be checked. Every reference was verified against its Crossref record.

Data and code availability. Four Python scripts, using the standard library or NumPy/SciPy, reproduce every number in this paper. They are archived with this manuscript at https://doi.org/10.5281/zenodo.23687032 (version 1.0.0-candidate) and in the repository https://github.com/ipitchford/initialisation-dependent-fixation (tag `v1.0.0-candidate`), whose README lists the commands. `consensus/certify_limits.py` gives the rational certificates of Appendix A. `consensus/evidence/verify_finite_n.py` solves the scalar finite-population chains and checks the rational chain (34). `consensus/evidence/multitype_bracket.py` integrates (48) for the bracket (53). `consensus/evidence/stationary_identity_and_caps.py` gives the exact certification of Theorem 6 and Corollary 7, the multitype finite-population chain of Table 4, and the rate-cap values of Table 3. No stochastic simulation is used.

Competing interests and funding. None.

Appendix A. Rational certificates

All certified decimals in this paper come from exact rational enclosures computed with Python's `fractions` module. For $x \geq 0$ the script encloses e^x between a Taylor polynomial of degree n and that polynomial plus a geometric tail bound, $x^{n+1}/((n+1)!\,(1 - x/(n+2)))$, valid for $x < n+2$.

The series (29) has positive terms with increasing denominators $\omega + j$. Its tail after 100 terms is therefore bounded by the next term divided by $1 - \zeta/102$.

These enclosures give rational bounds on c_F^{-1}, and hence on c_F. For the global supremum, the script brackets ξ_* by checking the sign of $e^{-\xi}(1+\xi)^3 - (\xi + e^{-\xi})^2$, which has the sign of Δ', at the two rational endpoints in (41). It bounds $\Delta(\xi_*)$ using the monotonicity of $\xi/(\xi + e^{-\xi})$ and $\xi/(1+\xi)$. The complete fractions, with numerators and denominators of several hundred digits, are stored in `certified_limits.json`. The certified decimal brackets are:

- constant-rate witness: the lower and upper rational bounds on c_F agree to 30 significant digits, $c_F = 0.702032053928212711530878678865\ldots$;
- second witness: likewise, $c_F = 0.987409482520935991411990627158\ldots$;
- maximiser: $\xi_* \in (1.71891151345837,\ 1.71891151345839)$;
- supremum: $\Delta(\xi_*) \in (0.273355434321512683887747,\ 0.273355434321518094791934)$.

For Theorem 6, the script brackets c^* to width 10^{-40} by rational bisection, using the strictly decreasing function $c \mapsto \int \lambda(x + \lambda c)^{-1} F_A(dx)$. Both fixation formulas are increasing in c, so the bracket gives certified enclosures. These contain $7/9$ and $(3 + \sqrt{37})/12$ for the example of Table 1.

References

Alexandre, A., Abbara, A., Fruet, C., Loverdo, C., & Bitbol, A.-F. (2025). Bridging Wright–Fisher and Moran models. *Journal of Theoretical Biology, 599*, 112030. https://doi.org/10.1016/j.jtbi.2024.112030

Athreya, K. B., & Ney, P. E. (1972). *Branching processes.* Springer. https://doi.org/10.1007/978-3-642-65371-1

Brown, M. (1980). Bounds, inequalities, and monotonicity properties for some specialized renewal processes. *The Annals of Probability, 8*(2), 227–240. https://doi.org/10.1214/aop/1176994773

Ethier, S. N., & Kurtz, T. G. (1986). *Markov processes: Characterization and convergence.* Wiley. https://doi.org/10.1002/9780470316658

Fernández-Gracia, J., Eguíluz, V. M., & San Miguel, M. (2011). Update rules and interevent time distributions: Slow ordering versus no ordering in the voter model. *Physical Review E, 84*(1), 015103. https://doi.org/10.1103/PhysRevE.84.015103

Ganesh, A., Hauert, S., & Valla, E. (2025). Consensus in the weighted voter model with noise-free and noisy observations. *Swarm Intelligence, 19*(3), 173–214. https://doi.org/10.1007/s11721-025-00248-z

Hubbarde, J. E., Wild, G., & Wahl, L. M. (2007). Fixation probabilities when generation times are variable: The burst–death model. *Genetics, 176*(3), 1703–1712. https://doi.org/10.1534/genetics.107.072009

Jafarpour, F., Levien, E., & Amir, A. (2023). Evolutionary dynamics in non-Markovian models of microbial populations. *Physical Review E, 108*(3), 034402. https://doi.org/10.1103/PhysRevE.108.034402

Karsai, M., Kivelä, M., Pan, R. K., Kaski, K., Kertész, J., Barabási, A.-L., & Saramäki, J. (2011). Small but slow world: How network topology and burstiness slow down spreading. *Physical Review E, 83*(2), 025102. https://doi.org/10.1103/PhysRevE.83.025102

Kendall, D. G. (1948). On the generalized "birth-and-death" process. *The Annals of Mathematical Statistics, 19*(1), 1–15. https://doi.org/10.1214/aoms/1177730285

Pal, A., Kostinski, S., & Reuveni, S. (2022). The inspection paradox in stochastic resetting. *Journal of Physics A: Mathematical and Theoretical, 55*(2), 021001. https://doi.org/10.1088/1751-8121/ac3cdf

Shpak, M. (2007). Selection against demographic stochasticity in age-structured populations. *Genetics, 177*(4), 2181–2194. https://doi.org/10.1534/genetics.107.080747

Stark, H.-U., Tessone, C. J., & Schweitzer, F. (2008). Decelerating microdynamics can accelerate macrodynamics in the voter model. *Physical Review Letters, 101*(1), 018701. https://doi.org/10.1103/PhysRevLett.101.018701

Uecker, H., & Hermisson, J. (2011). On the fixation process of a beneficial mutation in a variable environment. *Genetics, 188*(4), 915–930. https://doi.org/10.1534/genetics.110.124297

Valentini, G., Ferrante, E., & Dorigo, M. (2017). The best-of-n problem in robot swarms: Formalization, state of the art, and novel perspectives. *Frontiers in Robotics and AI*, 4, article 9. https://doi.org/10.3389/frobt.2017.00009

White, K. P. (1997). An effective truncation heuristic for bias reduction in simulation output. *Simulation, 69*(6), 323–334. https://doi.org/10.1177/003754979706900601